

Activity 2.6

Position and Speed Control Loop

Block name: Industrial Automation – MR2006B.602

Module 2: Microcontrollers

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Date: October 23, 2026

1 Introduction

This report covers the position and speed control of the car. First, the plant is identified from open-loop tests. Then a PID controller is designed, tuned and discretized in MATLAB/Simulink. Finally, it runs on the NUCLEO-F446RE. The car drives a target distance with a trapezoidal speed profile. It stops at obstacles closer than 30 mm and receives new parameters via Bluetooth.

2 Open-Loop Tests

The motors were driven with fixed PWM duty steps, without feedback. The encoders were sampled every 20 ms. Speed and distance were logged for each step.

Table 1: Open-loop step tests.

Duty [%]	**Final speed [mm/s]**	**Time to 63 % [s]**
20	[TBD]	[TBD]
40	[TBD]	[TBD]
60	[TBD]	[TBD]
80	[TBD]	[TBD]
100	650	[TBD]

The car starts moving at 14.1 % duty (left) and 16.9 % duty (right). Below these values, static friction holds the wheels.

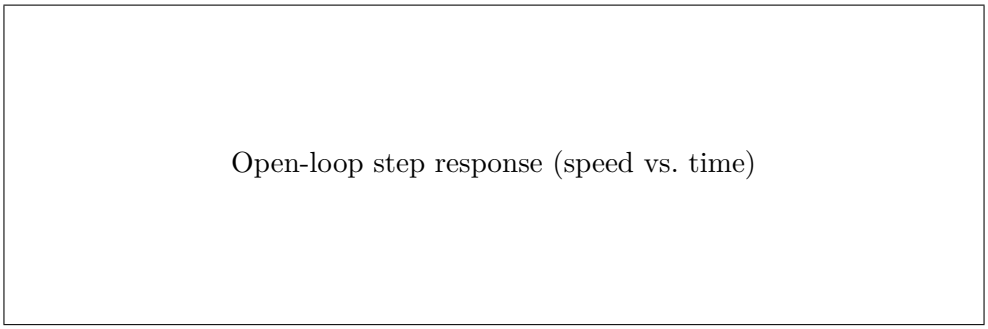


Figure 1: Open-loop step response.

3 Model Identification

Input u is the PWM duty in %. Output v is the wheel speed in mm/s. The MATLAB System Identification app estimated a first-order model:

$$G(s) = \frac{V(s)}{U(s)} = \frac{K}{\tau s + 1} \tag{1}$$

with $K =$ [TBD] mm/(s $\cdot$ %), $\tau =$ [TBD] s and a fit of [TBD] %. The position is the integral of the speed:

$$X(s) = \frac{G(s)}{s}\, U(s) \tag{2}$$

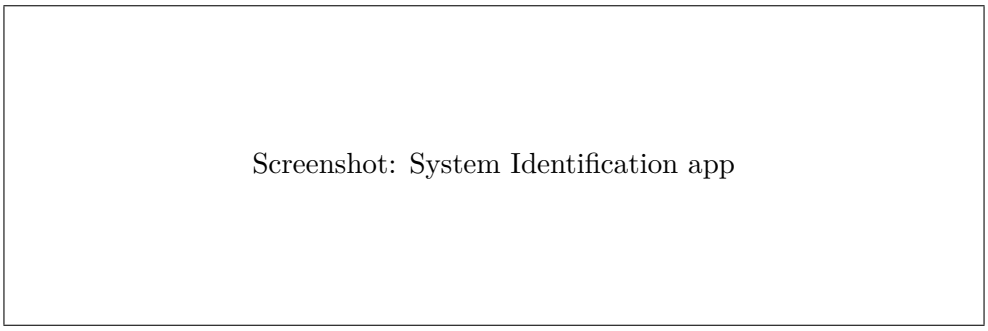


Figure 2: Model estimation in the System Identification app.

4 Control Loop Simulation

4.1 Closed Loop in Simulink

The loop has five blocks: reference generator, PID, actuator, plant and sensor.

- **Actuator:** H-bridge. Duty limited to $\pm 100\,\%$.
- **Plant:** $G(s)$ from the identification.
- **Sensor:** encoder with gain k_x, sampled every T_s.

The reference is a trapezoidal speed profile. The user sets the distance D, the maximum speed v_{max}, the acceleration time t_a and the deceleration time t_d:

$$v_{ref}(t) = \begin{cases} v_{max}\,\dfrac{t}{t_a} & 0 \le t < t_a \\ v_{max} & t_a \le t < T - t_d \\ v_{max}\,\dfrac{T-t}{t_d} & T - t_d \le t \le T \end{cases} \tag{3}$$

$$T = \frac{D}{v_{max}} + \frac{t_a + t_d}{2}, \qquad x_{ref}(t) = \int_0^t v_{ref}(\tau)\,d\tau \tag{4}$$

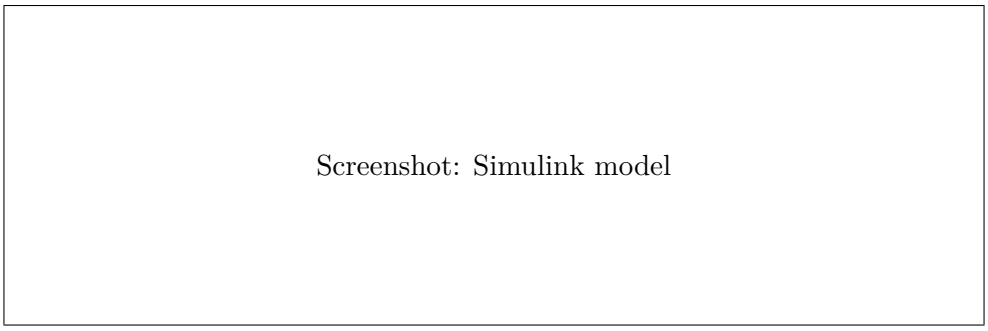


Figure 3: Closed control loop in Simulink.

4.2 Continuous PID

$$C(s) = K_p + \frac{K_i}{s} + K_d\,s \tag{5}$$

4.3 Discretization

The sampling time is $T_s = 20\,\text{ms}$. It is much smaller than τ (rule of thumb: $T_s \le \tau/10$). It also matches the TIM6 interrupt on the microcontroller. The plant was discretized with a zero-order hold. The discrete PID is:

$$C(z) = K_p + K_i\,T_s\,\frac{z}{z-1} + K_d\,\frac{z-1}{T_s\,z} \tag{6}$$

4.4 Tuning

The gains were tuned with the PID Tuner.

Table 2: PID gains.

	K_p	K_i	K_d
Continuous	[TBD]	[TBD]	[TBD]
Discrete	[TBD]	[TBD]	[TBD]

4.5 Validation

The continuous and the discrete loop give nearly the same response.

Table 3: Simulation results.

	Continuous	**Discrete**
Overshoot [%]	[TBD]	[TBD]
Settling time [s]	[TBD]	[TBD]
Final position error [mm]	[TBD]	[TBD]

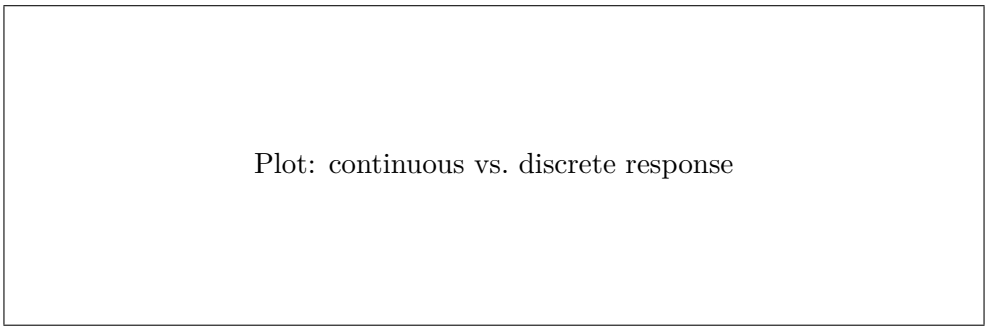


Figure 4: Continuous and discrete simulation.

5 Hardware

Table 4: Sensors, actuators and interfaces.

Device	**Interface**	**Pins**
2× BTS7960 H-bridge	PWM (TIM1)	PA8–PA11, PB12/13 (EN)
2× motor encoder	Encoder (TIM3/TIM8)	PA6, PA7; PC6, PC7
Ultrasonic sensor	GPIO (trig/echo)	[TBD]
OLED display	I^2C	[TBD]
Bluetooth module	UART	[TBD]

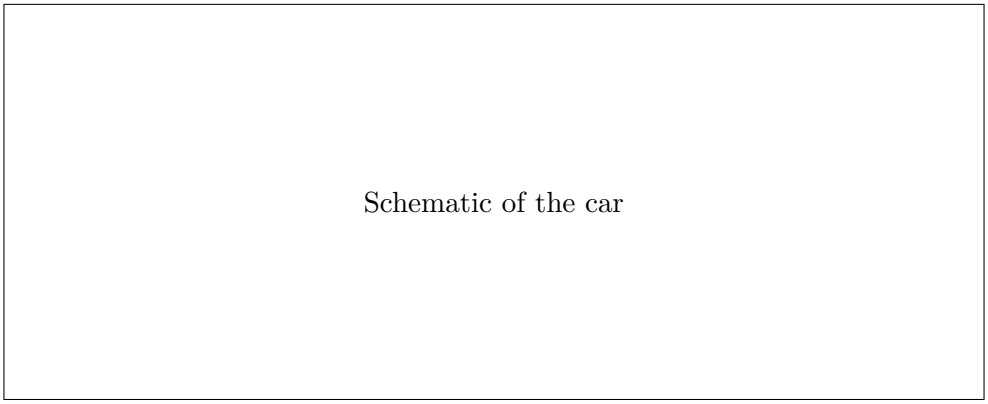


Figure 5: Wiring of the car.

6 Firmware Implementation

6.1 Sampling Period

TIM6 runs at $f_{TIM6} = 90\,\text{MHz}$ (APB1 $45\,\text{MHz} \times 2$). With PSC = 8999 and ARR = 199:

$$f_s = \frac{f_{TIM6}}{(PSC+1)(ARR+1)} = \frac{90 \cdot 10^6\,\text{Hz}}{9000 \cdot 200} = 50\,\text{Hz} \tag{7}$$

$$T_s = \frac{1}{f_s} = 20\,\text{ms} \tag{8}$$

6.2 PWM Signal

TIM1 generates the PWM for both H-bridges at 20 kHz.

$$f_{PWM} = \frac{f_{TIM1}}{(PSC+1)(ARR+1)} \tag{9}$$

$$(PSC+1)(ARR+1) = \frac{180 \cdot 10^6\,\text{Hz}}{20 \cdot 10^3\,\text{Hz}} = 9000 \tag{10}$$

$$\text{Duty} = \frac{CCR}{ARR+1} \cdot 100\,\% \tag{11}$$

Used values: PSC = [TBD], ARR = [TBD].

6.3 Distance and Speed

Each encoder gives $N_{rev} = 3584$ counts per wheel revolution (x4 decoding). The wheel circumference is $C_w \approx 228\,\text{mm}$.

$$k_x = \frac{C_w}{N_{rev}} = \frac{228}{3584} \approx 0.0636\,\text{mm/count} \tag{12}$$

$$x[k] = k_x\, N[k], \qquad v[k] = \frac{x[k] - x[k-1]}{T_s} \tag{13}$$

The speed resolution is $k_x / T_s \approx 3.18\,\text{mm/s}$.

6.4 PID Algorithm

The PID runs in the TIM6 callback, every T_s:

$$e[k] = r[k] - y[k] \tag{14}$$

$$I[k] = I[k-1] + K_i\, T_s\, e[k] \tag{15}$$

$$u[k] = K_p\, e[k] + I[k] + K_d\, \frac{e[k] - e[k-1]}{T_s} \tag{16}$$

The output is limited to $\pm 100\,\%$ duty. When the output saturates, the integral stops (anti-windup).

```
TIM6 callback (every 20 ms):
  read encoders            -> x, v
  next reference point     -> x_ref, v_ref
  if obstacle: hold reference, duty = 0
  else:        PID -> duty -> CCR (TIM1)
```

6.5 OLED Display

The OLED shows the state (run / stop / obstacle), the target distance, the current distance and speed, and the PID gains. It is refreshed in the main loop, outside the interrupt.

6.6 Bluetooth

Parameters are sent from a phone via the Bluetooth module (UART, [TBD] baud).

Table 5: Bluetooth commands.

Command	**Function**
`D<mm>`	Set target distance and start
[TBD]	Set K_p, K_i, K_d
[TBD]	Set v_{max}, t_a, t_d

6.7 Obstacle Detection

The ultrasonic sensor measures the echo time t_{echo}. With the speed of sound $c = 343\,\mathrm{m/s}$:

$$d = \frac{c\,t_{echo}}{2} \;\Rightarrow\; d\,[\mathrm{mm}] = 0.1715 \cdot t_{echo}\,[\mu\mathrm{s}] \tag{17}$$

If $d \leq 30\,\mathrm{mm}$, the motors stop and the reference is held. When the obstacle is removed, the car continues the profile.

7 Results

7.1 STM32CubeMonitor vs. MATLAB

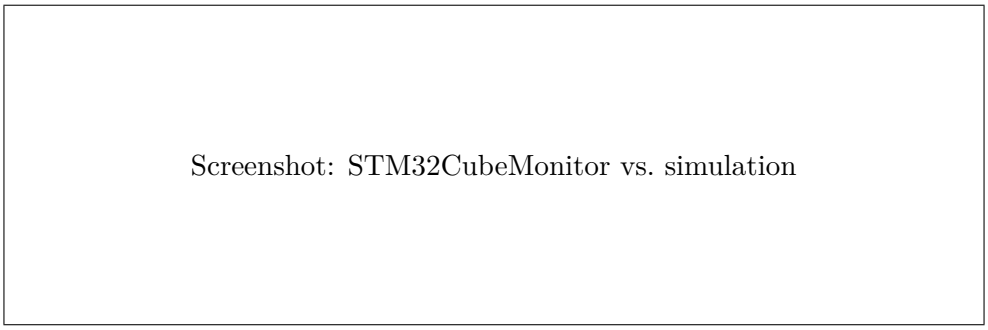


Figure 6: Measured and simulated response.

Table 6: Simulation and measurement.

	Simulation	**Measured**
Target distance [mm]	[TBD]	[TBD]
Distance traveled [mm]	[TBD]	[TBD]
Maximum speed [mm/s]	[TBD]	[TBD]
Travel time [s]	[TBD]	[TBD]

7.2 Obstacle Test

An obstacle was placed in front of the car during a move. The car stopped at [TBD] mm and continued after the obstacle was removed.

8 Conclusions

[TBD]